GAD+ Experiment Results

Transition State Search with Eckart-Projected GAD and HIP

April 2026

What This Is

We benchmark **Gentlest Ascent Dynamics (GAD)** for finding transition states using the HIP neural network potential on 300 organic reactions from Transition1x. GAD modifies forces to ascend along the lowest Hessian eigenvector while descending along all others: $\mathbf{F}_{\text{GAD}} = \mathbf{F} + 2(\mathbf{F} \cdot \mathbf{v}_1)\mathbf{v}_1$. A TS is converged when $n_{\text{neg}} = 1$ (one negative eigenvalue in the Eckart-projected vibrational Hessian) and force norm < 0.01 eV/Å. All methods are purely geometry-based—no path history. ∼15,000 total optimizations across 10 experiments.

1. Noise Robustness (300 samples, dt=0.01, 300 steps)

How does convergence degrade as we add Gaussian noise to the known TS geometry?

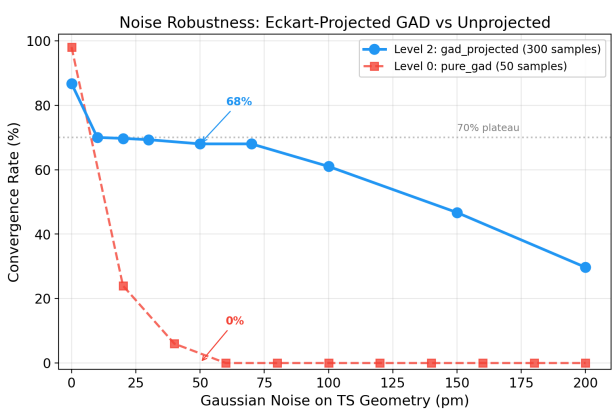

Figure 1: Level 2 (Eckart-projected, blue) vs Level 0 (unprojected, red). Eckart projection is worth +68pp at 50pm.

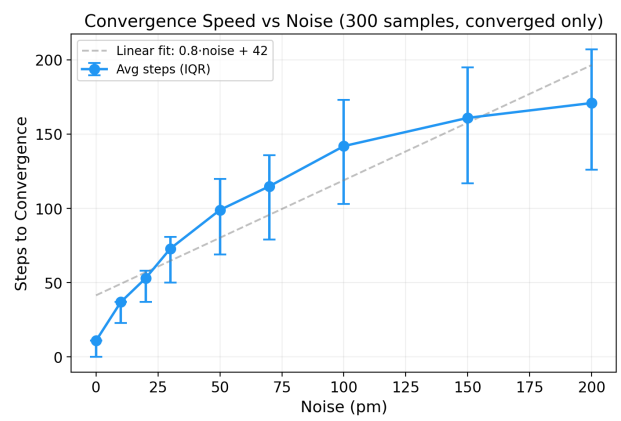

Figure 2: Steps to convergence scale linearly with noise. Error bars show IQR.

Noise	Conv	Rate	Steps	Noise	Conv	Rate	Steps
0pm	260/300	86.7%	11	70pm	204/300	68.0%	115
10pm	210/300	70.0%	37	100pm	183/300	61.0%	142
20pm	209/300	69.7%	53	150pm	140/300	46.7%	161
50pm	204/300	68.0%	99	200pm	89/300	29.7%	171

Takeaway: ∼70% convergence plateau from 10–70pm. Level 0 (no projection) gets 0% at ≥50pm. Eckart projection is the single largest improvement.

2. Starting Geometry (300 samples, dt=0.01, 300 steps)

Can GAD find a TS from geometries other than a noised TS?

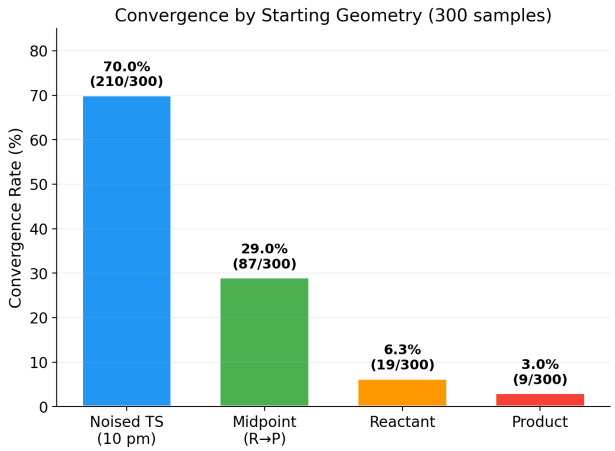

Figure 3: GAD needs saddle-region starts. Reactant/product equilibria have $n_{\text{neg}} = 0$, so GAD has no direction to ascend.

Start	Conv	Rate	Steps
Noised TS	210/300	70.0%	37
Midpoint R→P	87/300	29.0%	191
Reactant	19/300	6.3%	108
Product	9/300	3.0%	65

Takeaway: Midpoint interpolation is a viable starting point (29%) without knowing the TS. Equilibrium geometries are near-useless.

3. Basin of Attraction (50 samples, dt=0.01, 300 steps)

At what noise level does GAD start finding a *different* TS?

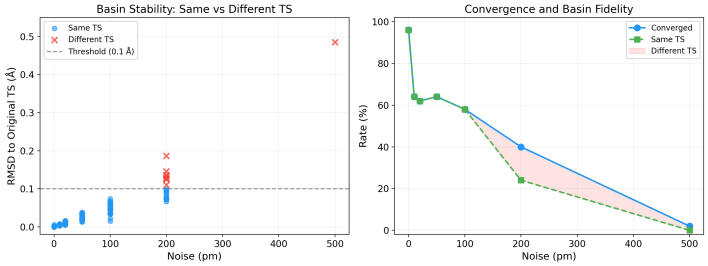

Figure 4: Left: RMSD to original TS (blue = same, red = different). **Right:** Same-TS rate vs noise.

Noise	Conv	Same	Diff
0pm	48/50	48	0
10pm	32/50	32	0
50pm	32/50	32	0
100pm	29/50	29	0
200pm	20/50	12	8
500pm	1/50	0	1

Takeaway: Basin stable to 100pm—**zero wrong TS found** in 172 converged runs. GAD is either right or fails. No silent errors.

4. 7-Method Comparison (300 samples, 1000 steps, 12,600 runs)

Which GAD variant works best? 42 parallel MIG jobs.

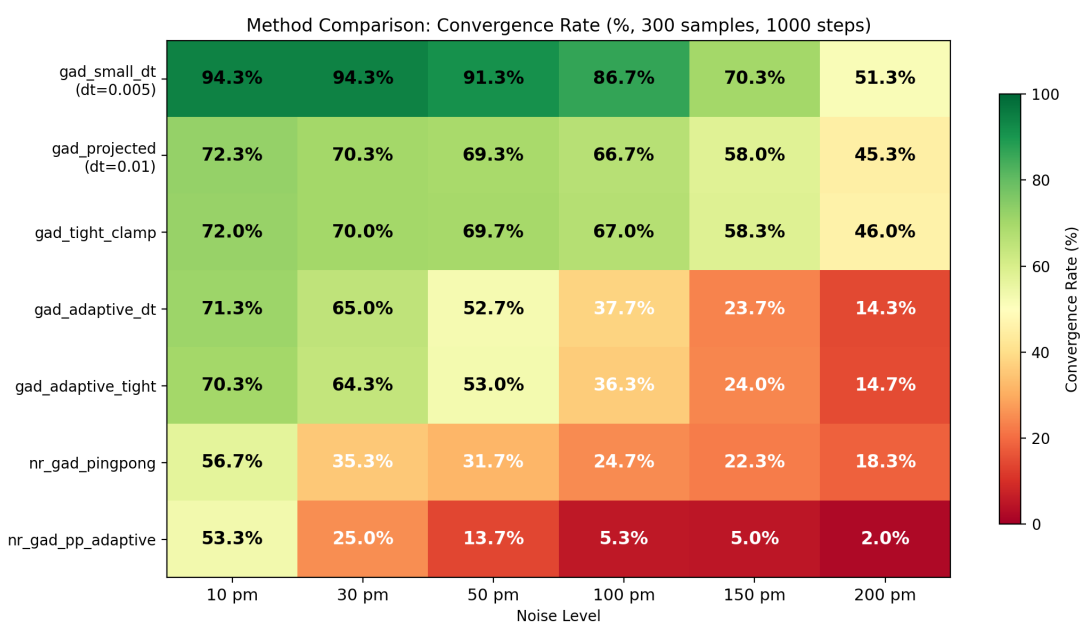

Figure 5: Convergence rate heatmap. `gad_small_dt` (dt=0.005) dominates everywhere. Adaptive dt and NR-GAD ping-pong both hurt.

Method	10pm	30pm	50pm	100pm	150pm	200pm
gad_small_dt (dt=0.005)	**94.3**	**94.3**	**91.3**	**86.7**	**70.3**	**51.3**
gad_projected (dt=0.01)	72.3	70.3	69.3	66.7	58.0	45.3
gad_tight_clamp	72.0	70.0	69.7	67.0	58.3	46.0
gad_adaptive_dt	71.3	65.0	52.7	37.7	23.7	14.3
nr_gad_pingpong	56.7	35.3	31.7	24.7	22.3	18.3
nr_gad_pp_adaptive	53.3	25.0	13.7	5.3	5.0	2.0

Takeaways:

- **dt=0.005 beats dt=0.01** by 20+pp at all noise levels. Smaller steps avoid overshooting the narrow saddle region where n_{neg} transitions 2→1.
- **Adaptive dt hurts.** Eigenvalue-clamped strategy makes steps too small near saddle points where large $|\lambda_1|$ triggers excessive shrinking.
- **NR-GAD ping-pong hurts.** Newton descent when $n_{\text{neg}} \geq 2$ overshoots even with damping. Pure GAD already handles the $n_{\text{neg}} \geq 2 \to 1$ transition.
- **Tight clamping does nothing.** Default 0.35Å cap never triggers.

5. Damped NR-GAD (300 samples, 1000 steps)

Can step damping ($\alpha \cdot \Delta\mathbf{x}$, norm cap) fix NR-GAD?

Method	10pm	30pm	50pm	100pm	150pm	200pm
gad_small_dt	**94.3**	**94.3**	**91.3**	**86.7**	**70.3**	**51.3**
damped α=0.1	94.7	88.0	77.7	58.0	46.0	33.7
damped α=0.2	93.0	88.0	78.3	60.3	47.3	36.3
damped α=0.3	88.7	82.3	75.0	58.7	46.3	37.0

Takeaway: Damping helps at 10pm (α=0.1 matches baseline) but hurts everywhere else. Pure GAD is strictly better. The NR step is redundant—GAD's own dynamics already navigate $n_{\text{neg}} \geq 2 \to 1$ correctly.

6. Step Budget Ceiling (300 samples, dt=0.005)

Does more compute push convergence higher?

Noise	1000 steps	2000 steps	3000 steps
10pm	94.3%	—	95.7% (+1.4)
50pm	91.3%	—	93.0% (+1.7)
100pm	86.7%	—	88.7% (+2.0)
150pm	70.3%	74.0% (+3.7)	timed out

Takeaway: Ceiling is ∼96%. 3× the compute gives only 1–2pp. The remaining ∼4% are structurally hard molecules.

7. Trajectory Analysis

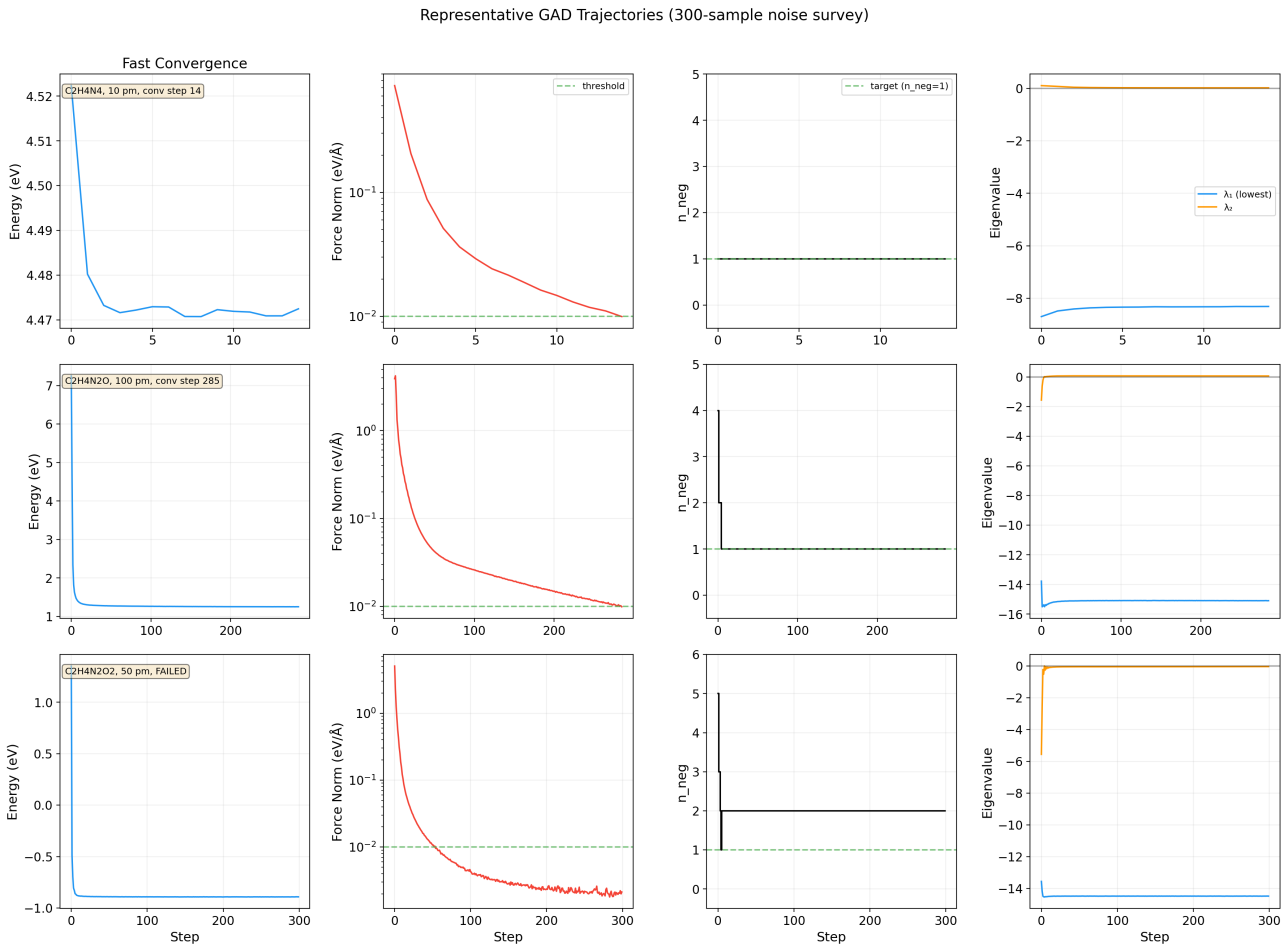

Figure 6: Representative trajectories: fast convergence (top), slow convergence (middle), failure (bottom). Columns: energy, force norm, n_{neg}, eigenvalues.

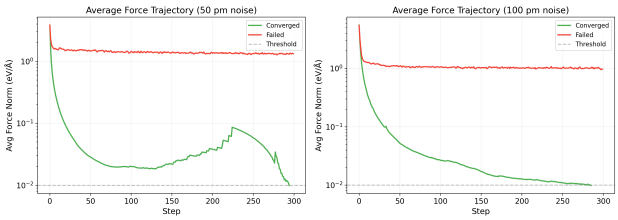

Figure 7: Average force: converged (green) drops exponentially, failed (red) plateaus.

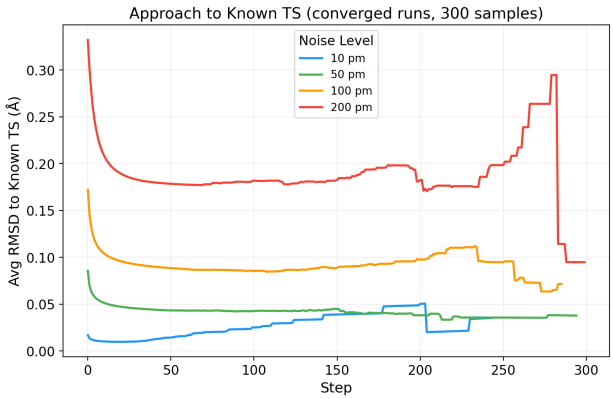

Figure 8: RMSD to known TS decreases monotonically. Higher noise starts farther.

8. IRC Validation (30 samples × 3 noise levels, threshold 0.5Å)

When GAD converges, is it the *intended* TS?

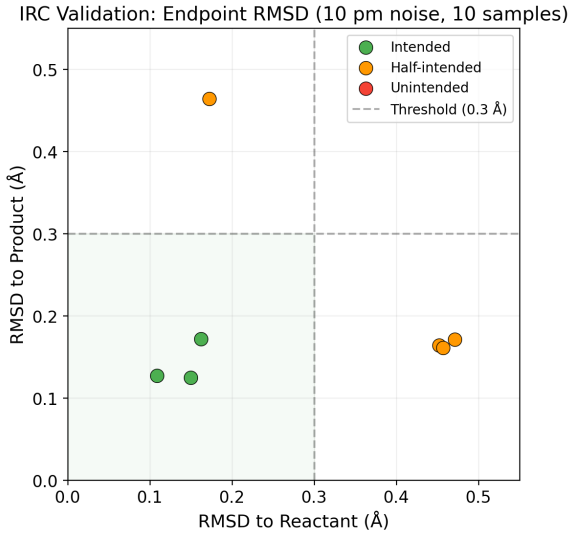

Figure 9: IRC endpoint RMSD. Green region = intended.

Noise	Intended	Half	Unintended
10pm	17/30 (57%)	3 (10%)	10 (33%)
50pm	20/30 (67%)	3 (10%)	7 (23%)
100pm	19/30 (63%)	4 (13%)	7 (23%)

Takeaway: 57–67% intended. Noise *increases* intended rate (57%→67% from 10→50pm)—may act as regularizer helping escape non-target saddle points.

Summary

Finding	Evidence
Eckart projection is everything	0% → 68% at 50pm
dt=0.005 is optimal	94% at 10pm, 91% at 50pm, 87% at 100pm
Basin stable to 100pm	0 wrong TS in 172 converged runs
Ceiling ∼96%	3000 steps gains only 1–2pp
IRC 57–67% intended	Noise helps (regularization)
Adaptive dt hurts	−20–30pp vs baseline
NR-GAD hurts (even damped)	Pure GAD is strictly better
Tight clamping: no effect	0.1Å vs 0.35Å cap, <1pp
Midpoint start viable	29% without knowing TS

Best configuration: `gad_small_dt` — Eckart-projected GAD, dt=0.005, 1000 steps, no mode tracking, no adaptive dt, no NR. The simplest method wins.